

Roll No.

TBT-602

B. TECH. (BIOTECHNOLOGY) (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, March, 2024

PLANT BIOTECHNOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What is culture media ? Explain the components of MS media. (CO1)

OR

(b) Write the principle of Plant Tissue Culture. Explain the stages of PTC.

(CO1)

2. (a) Write advantages and disadvantages of Plant Tissue Culture.

(CO1, CO2)

OR

(b) What is somaclonal variation ? Explain the molecular basis of somaclonal variation.

(CO1, CO2)

P. T. O.

(2)

3. (a) Explain the sterilization techniques used in PTC. (CO1, CO2)

OR

- (b) Write short notes on the following : (CO1, CO2)

(i) Clonal Propagation

(ii) Callus culture

4. (a) What do you understand by somatic embryogenesis ? Explain its application in Plant tissue culture. (CO1, CO2)

OR

- (b) Discuss the role of PTC in conservation of threatened plant species.

(CO1, CO2)

5. (a) What is cryopreservation ? Explain the method of cryopreservation.

(CO2, CO3)

OR

- (b) Explain the method of synthetic seed production. How are synthetic seeds different from natural seed ? (CO2, CO3)

Roll No.

TBT-601

**B. TECH. (BIOTECHNOLOGY) (SIXTH SEMESTER)
MID SEMESTER EXAMINATION, March, 2024**

ANIMAL BIOTECHNOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What is the function of CO₂ and water in animal cell culture incubator ?
Describe the functions of these factors in detail. (CO1)

OR

- (b) What is primary cell ? Provide different methods of primary cell culture preparation. (CO1)

2. (a) Describe the functions and components of serum ? Why quality control of serum is of paramount importance ? Explain it. (CO1)

OR

- (b) What are major contaminants in animal cell culture ? Describe various antibiotics used for bacterial, fungal and mycoplasma infection. (CO1)

3. (a) What are the major considerations in animal cell culture laboratory set up ? Describe in detail. (CO1)

P. T. O.

(2)

OR

(b) Describe : (CO1)

(i) Complement fixation

(ii) Cell culture medium

4. (a) What are retroviral vectors ? Describe their types and important features in detail. (CO2)

OR

(b) What is YAC ? Describe its structure and important functions in detail. (CO2)

5. (a) Describe major methods of transgenic animal production along with few examples. (CO2)

OR

(b) What is IVF ? Explain detailed procedure of IVF in humans. (CO2)

Roll No.

TBT-603

B. TECH. (BIOTECHNOLOGY) (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, March, 2024

BIOSEPARATION TECHNOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Describe various classes of Bioproducts and their specific examples with applications. (CO1)

OR

- (b) What are *five* stages in the downstream processing of Bioproducts ?
What are the problems faced in bioseparations ? (CO1)

2. (a) Give the table showing the factors and specific unit operation needed for downstream process operation. (CO1)

OR

- (b) Explain the downstream processing for Ethanol and any intracellular enzyme using flowcharts. (CO1)

P. T. O.

(2)

3. (a) Describe chemical and enzymatic methods of cell disruption in detail.

(CO2)

OR

- (b) Write a short note on Disc Bowl Centrifuges and Centrifugal Force.

(CO2)

4. (a) Explain the principle of solvent extraction and problems associated with solvent extraction.

(CO2)

OR

- (b) Describe Plate and Frame Filter Press. What is Poiseuille's equation and its significance in filtration ?

(CO2)

5. (a) Discuss the following :

(CO2)

(i) Salting out using Ammonium Sulphate

(ii) Isoelectric precipitation and Cohn's equation

OR

- (b) Explain Physical and Dissociative Extraction processes. How are they different ?

(CO2)

Roll No.

TBT-604(a)

B. TECH. (BIOTECHNOLOGY) (SIXTH SEMESTER) MID SEMESTER EXAMINATION, March, 2024

BIOSIMILAR TECHNOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) What are the differences between a generic and a biosimilar ? (CO1)

OR

- (b) Explain chemical genomics approaches.

2. (a) Describe the goal, scope, structure and components of biosimilar technology regulatory guideline of India. (CO1)

OR

- (b) "Biosimilar market growth driven by increasing development of biologics." Comment. (CO1)

3. (a) Describe the prerequisites for biosimilar development. (CO1)

OR

- (b) Explain how researchers ensure the success of biosimilar on which they are working. (CO1)

P. T. O.

(2)

4. (a) What is gene cell therapy ? Describe its *two* examples with indications for which they were developed. (CO2)

OR

- (b) Explain stem cell biologics by giving *two* examples. (CO2)

5. (a) Write about Gardasil, Zostavax and Spikevax. (CO2)

OR

- (b) Write about Rituximab, Ustekinumab and Eculizumab. (CO2)

Roll No.

TBT-604(c)

B. TECH. (BIOTECHNOLOGY) (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, March, 2024

MICROBIAL TECHNOLOGY

Time : 1½ Hours

Maximum Marks : 50

- Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Discuss the role of microorganisms in medical, agriculture and food and beverage industries. (CO1)

OR

- (b) Define biopolymers and explain its production. (CO1)

2. (a) Implement the knowledge of recombinant DNA technology in production of hGH. (CO3, CO1)

OR

- (b) Differentiate between Gram-positive and Gram-negative bacteria with examples. (CO1, CO2)

P. T. O.

(2)

3. (a) Discuss synthetic and DNA vaccines. (CO1, CO2)

OR

- (b) Discuss the general principle of production of recombinant protein.

(CO1, CO2)

4. (a) Draw the well labelled diagram of Hepatitis-B virus and give diagrammatic representation of its vaccine production. (CO2)

OR

- (b) Discuss the production of Xanthan gum with its applications. (CO2)

5. (a) Explain the production of recombinant insulin. (CO1)

OR

- (b) Define the term gene expression. Define the process of RNA splicing takes place after transcription process in eukaryotes. (CO1)